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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-45

Transformer Design For Isolated DC-DC Converter-II

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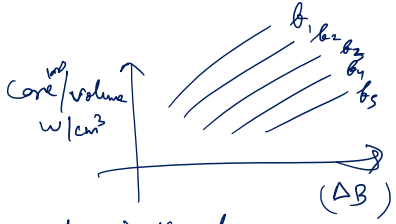
Recap

- Basic principles of transformer and losses associated with it.
- Design Outcome
  - Core size (core cross-sectional area, window area, mean length per turn, magnetic path length)
  - Number of turns
  - Wire size
  - Peak flux density
- Core loss

$$P_{core} = K_{core}(\Delta B)^{\beta} A_c l_c$$

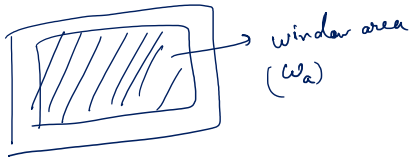
- Maximum flux density

$$\Delta B = \frac{\lambda_1}{2n_1 A_c}$$



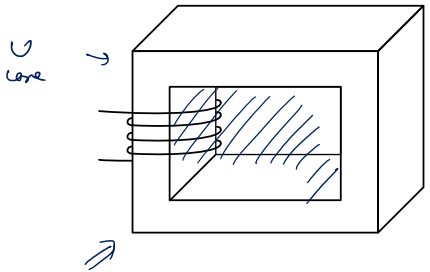
Copper Loss

- Transformer has multiple windings, wound over the core
- Windings (assume *k* windings) need to be accommodated in the entire window area
- The fraction (γ_j) of the window area allotted to winding *j*



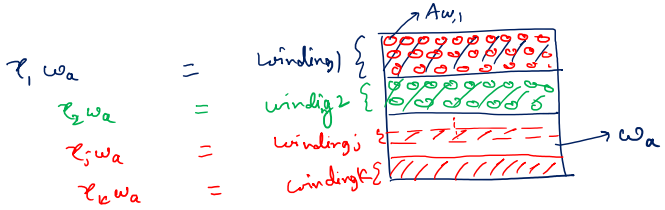
$$0 < \gamma_j < 1 \qquad (j = 1, 2, 3 \cdots k)$$

$$\gamma_1 + \gamma_2 + \gamma_3 \ldots\ldots + \gamma_k = 1$$



$\gamma_1 W_a + \gamma_2 W_a + \cdots \gamma_k W_a = W_a$

$\Rightarrow \gamma_1 + \gamma_2 + \gamma_3 + \cdots \gamma_k = 1$



Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Copper Loss

- Windings have copper loss → $I^2_{rms} R$
- The copper loss $P_{cu,j}$ in j^{th} winding depends on the dc resistance R_j of winding

$$P_{cu,j} = I_j^2 R_j \rightarrow \text{rms}$$

While,

$$R_j = \rho \frac{l_j}{A_{w,j}} \rightarrow \text{neglect the high freq. AC effects (Skin \& Proximity effect)}$$

Where

- ρ is the wire resistivity
- l_j is the length of the wire used for winding j
- $A_{w,j}$ is the cross-sectional area of the wire used for winding j
- I_j is the RMS value of the winding j current

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Copper Loss Cont.

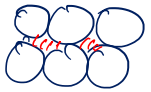
Length and cross sectional area of wire used in winding j to be

$l_j = n_j (MLT)$ → Mean length per turn; → no. of turns of the jth winding; → length of wire used for j^{th} winding



$A_{W,j} = \frac{W_a K_j K_u}{n_j}$ → window area; → fraction window area allotted to j^{th} winding; → no. of turns of jth winding

↓ area of cross section of wire in j^{th} winding



K_u → winding fill factor < 1

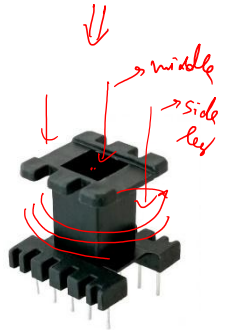
$K_u = 0.5$ for LV round-wire conductor

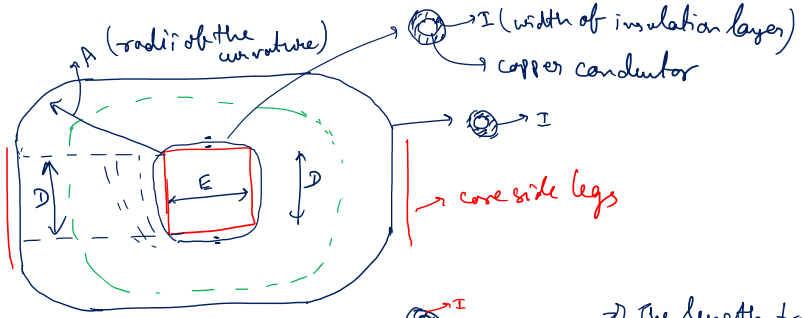
$K_u = 0.65$ for LV foil winding

$K_u = 0.1$ for HV transformer winding

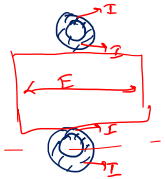
Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Copper Loss Cont.





$$MLT = \pi A + \left[\frac{(2D + 4I) + (2E + 4I)}{2} \times 2\right]$$



⇒ The length transverse by the curvature part = $\frac{2\pi(A) \times 4}{4}$

⇒ Mean length transverse by the curvature part = $2\pi A/2 = \pi A$

⇒ PQ, U

Source: https://royalcomponents.in/index.php?route=product/product&product_id=501361837727

Copper Loss Cont.

⇒ Resistance of jth winding, $R_j = \frac{\rho l_j}{A_{w,j}}$

$l_j = n_j (MLT)$

$A_{w,j} = \frac{W_a K_u x_j}{n_j}$

⇒ $R_j = \frac{\rho n_j^2 (MLT)}{W_a K_u x_j}$ → (neglecting the AC effect)

⇒ The copper loss of the jth winding

$P_{w,j} = \frac{n_j^2 I_j^2 \rho (MLT)}{W_a K_u x_j}$

(I_j → the RMS current rating of the jth winding)

⇒ The total copper loss of the 'k' windings

$P_{w,tot} = P_{w,1} + P_{w,2} + \cdots P_{w,k}$

$= \frac{\rho (MLT)}{W_a K_u} \sum_{j=1}^{k} \left(\frac{n_j^2 I_j^2}{x_j} \right)$

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Copper Loss Cont.

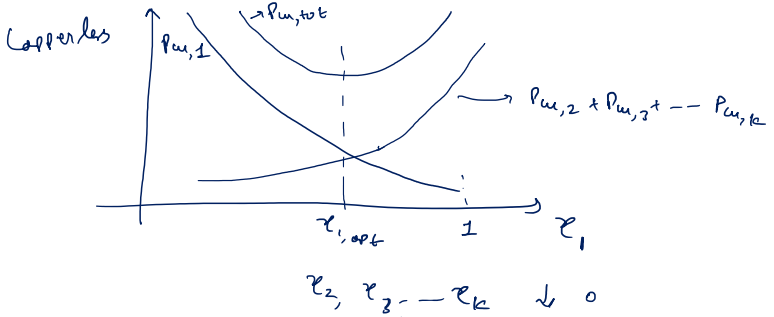
$$P_{w,tot} = \frac{\rho (MLT)}{w_a k_w} n_1^2 \sum_{j=1}^{k} \left(\frac{n_j'}{n_1}\right)^2 \frac{I_j^2}{x_j}$$

$$\left(n_j' = \left(\frac{n_j'}{n_1}\right) \times n_1 \right)$$ turns ratio → specification

$$n_1 = \frac{d_1}{2 \Delta B A_c}$$

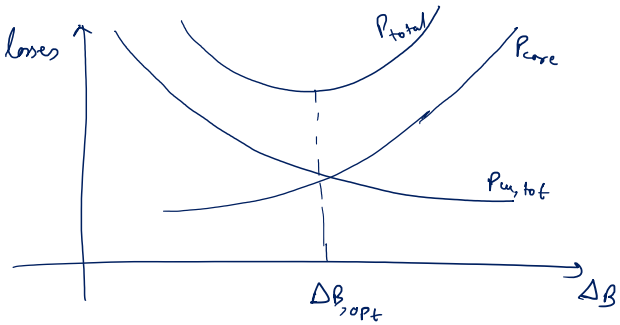
$$\Rightarrow P_{w,tot} = \frac{\rho (MLT)}{w_a k_w} \times \frac{d_1^2}{4 (\Delta B)^2 \cdot A_c} \sum_{j=1}^{k} \left(\frac{n_j'}{n_1}\right)^2 \cdot \frac{I_j^2}{x_j}$$

$$\Rightarrow P_{w,tot} \propto \left(\frac{1}{\Delta B}\right)^2$$



Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Total Loss



$\Rightarrow$ Total Power loss, $P_{total} = P_{core} + P_{w,tot}$

$P_{core} \propto (\Delta B)^{\beta}$ ($\beta \rightarrow$ 2.5 to 2.7)

$P_{w,tot} \propto \left(\frac{1}{\Delta B}\right)^2$

we need to find optimum value of ΔB for which P_{total} is minimum

$\Rightarrow \frac{d(P_{total})}{d(\Delta B)} = 0$

$\Rightarrow \frac{d(P_{core})}{d(\Delta B)} = -\frac{d(P_{w,tot})}{d(\Delta B)} \Rightarrow \Delta B_{,opt}$

This $\Delta B_{,opt}$ is used to obtained the $P_{total,min}$ (minimum value of total loss)

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

(1) $P_{core} = K_{core} (\Delta B)^{\beta} A_c l_c$

(2) $P_{w,tot} \propto \left(\frac{1}{\Delta B}\right)^2$

(3) $\Delta B = \frac{d_1}{2 n_1 A_c}$

$\Rightarrow$ $x_{1,opt}, x_{2,opt}, \cdots x_{k,opt}$ $\Rightarrow$ min. copper loss in the transformer winding

$\Rightarrow$ ΔB_{opt} $\Rightarrow$ min. total loss in the transformer

Source: R. W. Erickson and D. Maksimovic, Fundamentals of Power Electronics, Springer, 2nd Ed., 2001

Thank You